

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('gender_submission.csv')
df1 = pd.read_csv('test.csv')
df2 = pd.read_csv('train.csv')
```

df



	PassengerId	Survived	
0	892	0	
1	893	1	
2	894	0	
3	895	0	
4	896	1	
...	...	...	
413	1305	0	
414	1306	1	
415	1307	0	
416	1308	0	
417	1309	0	

418 rows × 2 columns

Next steps:

[Generate code with df](#)

[View recommended plots](#)

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df1



	PassengerId	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked	
0	892	3	Kelly, Mr. James	male	34.5	0	0	330911	7.8292	NaN	Q	
1	893	3	Wilkes, Mrs. James (Ellen Needs)	female	47.0	1	0	363272	7.0000	NaN	S	
2	894	2	Myles, Mr. Thomas Francis	male	62.0	0	0	240276	9.6875	NaN	Q	
3	895	3	Wirz, Mr. Albert	male	27.0	0	0	315154	8.6625	NaN	S	
4	896	3	Hirvonen, Mrs. Alexander (Helga E Lindqvist)	female	22.0	1	1	3101298	12.2875	NaN	S	
...	...	...	...	...	...	...	...	...	...	...	...	
413	1305	3	Spector, Mr. Woolf	male	NaN	0	0	A.5. 3236	8.0500	NaN	S	
414	1306	1	Oliva y Ocana, Dona. Fermina	female	39.0	0	0	PC 17758	108.9000	C105	C	
415	1307	3	Saether, Mr. Simon Sivertsen	male	38.5	0	0	SOTON/O.Q. 3101262	7.2500	NaN	S	
416	1308	3	Ware, Mr. Frederick	male	NaN	0	0	359309	8.0500	NaN	S	
417	1309	3	Peter, Master. Michael J	male	NaN	1	1	2668	22.3583	NaN	C	

Next steps:

[Generate code with df1](#)

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df2



	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S
...	...	...	...	...	...	...	...	...	...	...	...	...
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.0000	NaN	S
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.0000	B42	S
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.4500	NaN	S
889	890	1	1	Robb, Mr. Karl Howell	male	26.0	0	0	111360	30.0000	C148	C

Next steps: [Generate code with df2](#) [View recommended plots](#) [New interactive sheet](#)

df2.head()



	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S

Next steps: [Generate code with df2](#) [View recommended plots](#) [New interactive sheet](#)

```
df2.info()
df2.describe()
df2.isnull().sum()
df2['Survived'].value_counts()
```



```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  891 non-null    int64
1   Survived     891 non-null    int64
2   Pclass       891 non-null    int64
3   Name         891 non-null    object
4   Sex          891 non-null    object
5   Age          714 non-null    float64
6   SibSp        891 non-null    int64
7   Parch        891 non-null    int64
8   Ticket       891 non-null    object
9   Fare         891 non-null    float64
10  Cabin        204 non-null    object
11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

count

Survived

0	549
1	342

dtype: int64

```
df2['Pclass'].value_counts()
```

```

↵
count
Pclass
3      491
1      216
2      184

dtype: int64

```

```
df2['Sex'].value_counts()
```

```

↵
count
Sex
male    577
female  314

dtype: int64

```

```

#VISUAL EXPLORATION
#Univariate distribution

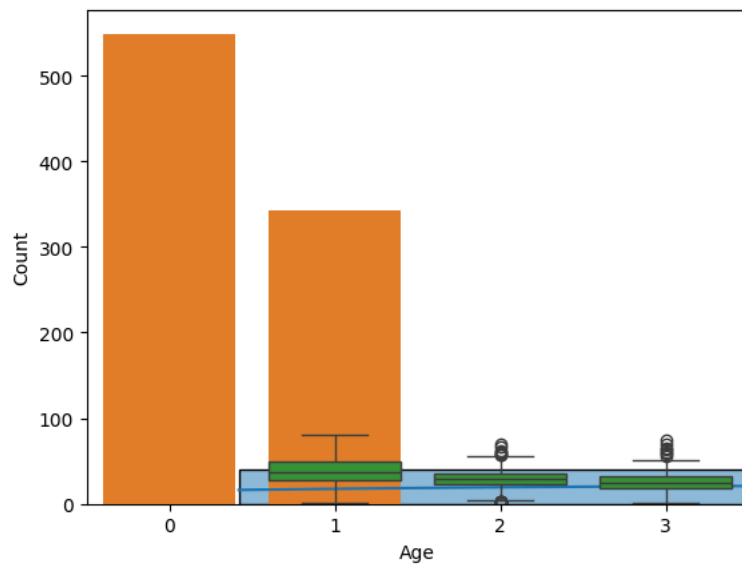
```

```

sns.histplot(df2['Age'].dropna(), kde=True)
sns.countplot(x='Survived', data=df2)
sns.boxplot(x='Pclass', y='Age', data=df2)

```

```
↵ <Axes: xlabel='Age', ylabel='Count'>
```



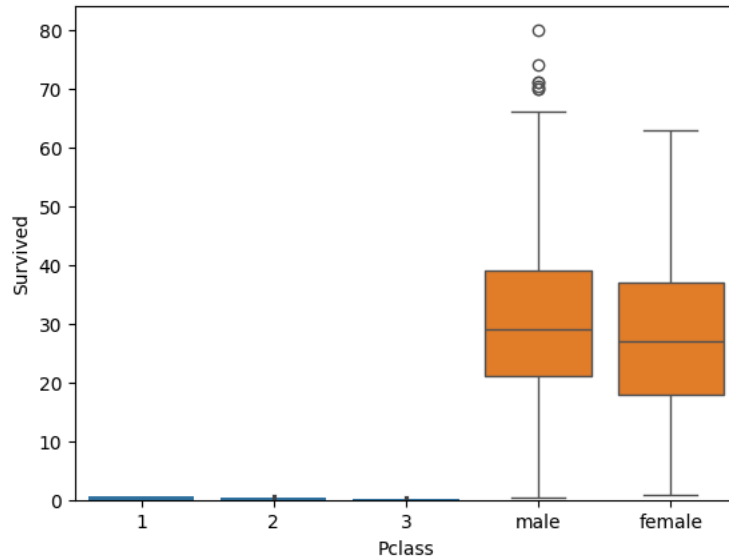
```
#Bivariate Relationships
```

```

sns.barplot(x='Pclass', y='Survived', data=df2)
sns.boxplot(x='Sex', y='Age', data=df2)

```

<Axes: xlabel='Pclass', ylabel='Survived'>

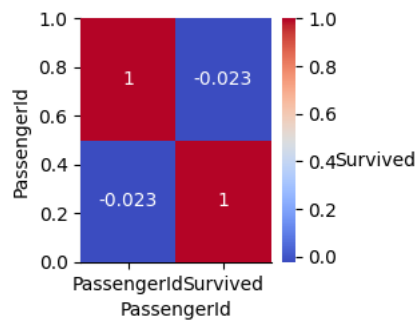


```
numeric_cols = df.select_dtypes(include=['float64', 'int64'])
```

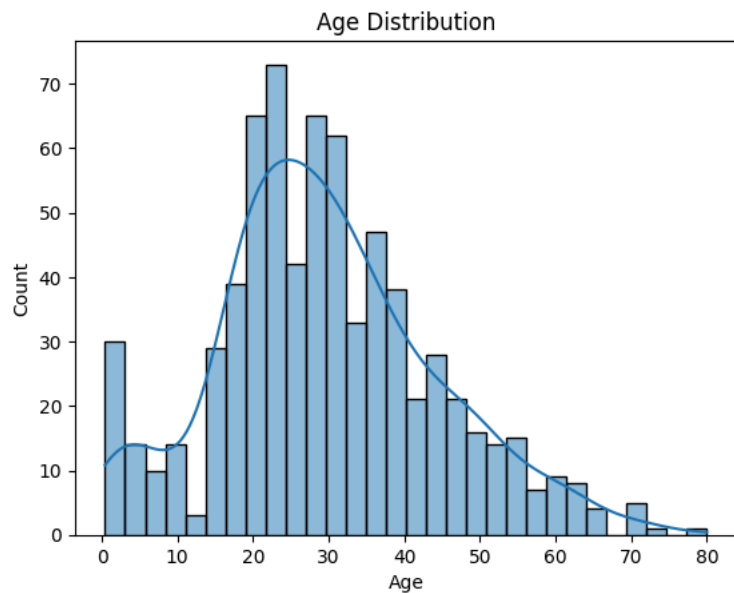
```
# Dropping rows with missing values in these columns for pairplot
sns.pairplot(numeric_cols.dropna(), hue='Survived')
```

```
# Heatmap of correlation between numeric variables
sns.heatmap(numeric_cols.corr(), annot=True, cmap='coolwarm')
```

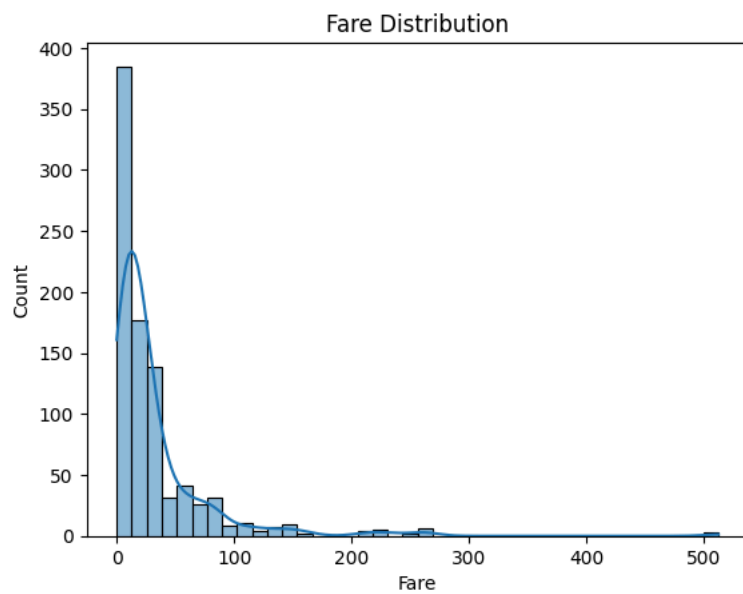
<Axes: >



```
#histogram
#age distribution
sns.histplot(df2['Age'].dropna(), bins=30, kde=True)
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Count")
plt.show()
```



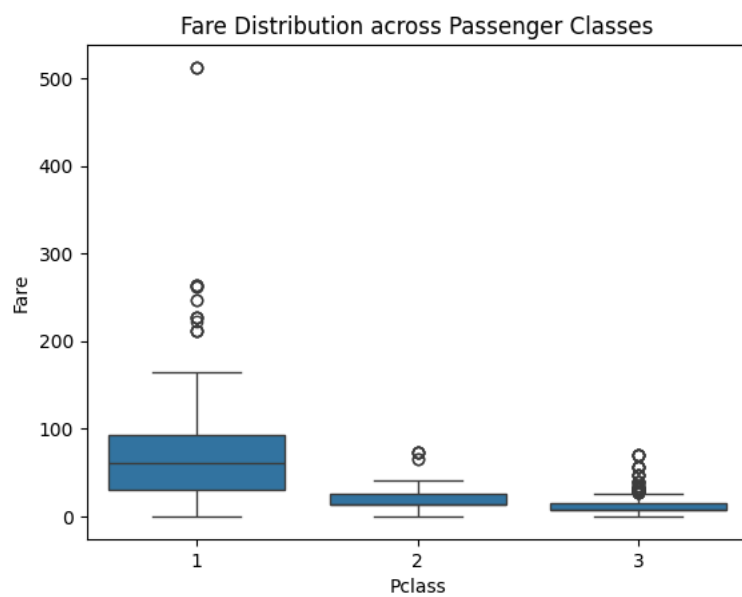
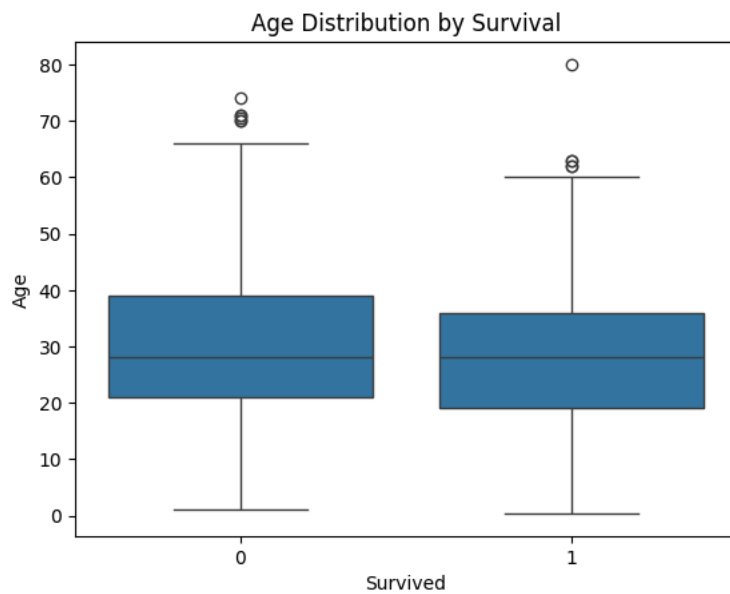
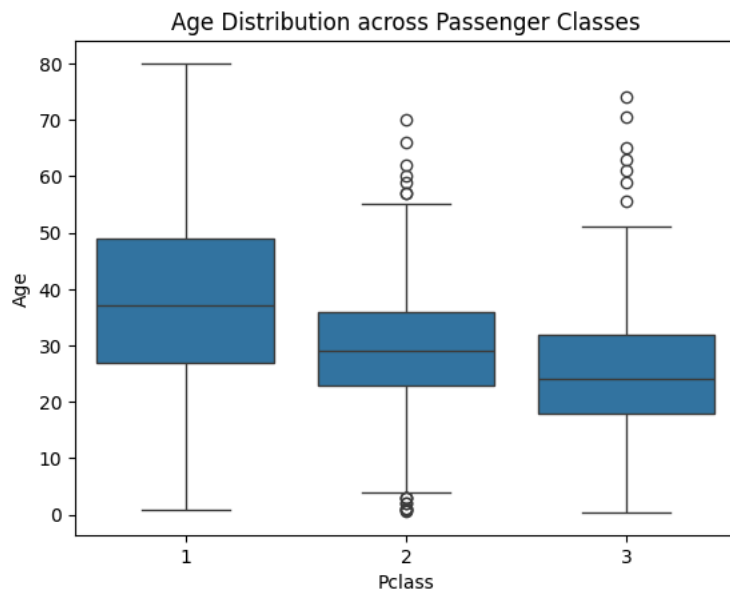
```
# Fare distribution
sns.histplot(df2['Fare'], bins=40, kde=True)
plt.title("Fare Distribution")
plt.xlabel("Fare")
plt.ylabel("Count")
plt.show()
```



```
#BOXPLOTS
# Age vs Pclass
sns.boxplot(x='Pclass', y='Age', data=df2)
plt.title("Age Distribution across Passenger Classes")
plt.show()

# Age vs Survived
sns.boxplot(x='Survived', y='Age', data=df2)
plt.title("Age Distribution by Survival")
plt.show()

# Fare vs Pclass
sns.boxplot(x='Pclass', y='Fare', data=df2)
plt.title("Fare Distribution across Passenger Classes")
plt.show()
```



```
#SCATTERPLOTS
```

```
# Age vs Fare colored by Survival
```

```
sns.scatterplot(x='Age', y='Fare', hue='Survived', data=df2)
```

```
plt.title("Fare vs Age by Survival")
```

```
plt.show()
```

```
# Parch vs Fare
```

```
sns.scatterplot(x='Parch', y='Fare', hue='Survived', data=df2)
```